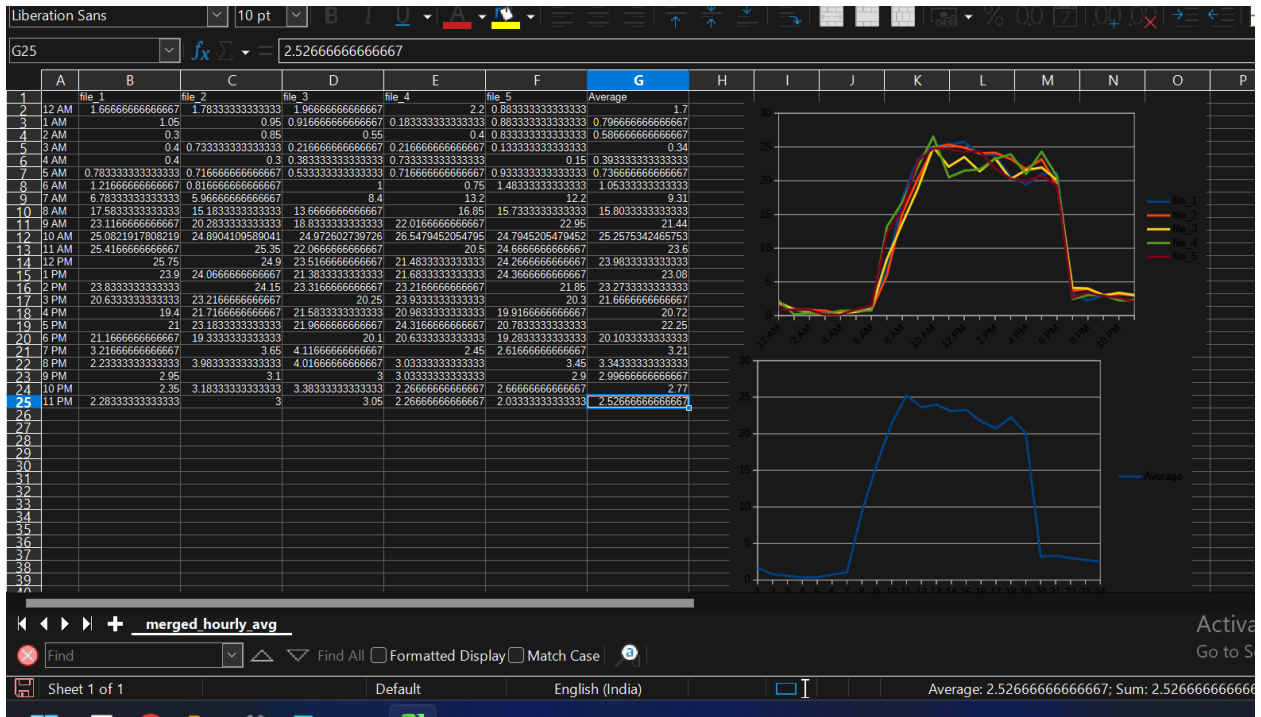


FREEDOM PARK 1.1 (Colab)

Look at the KR Circle fix data collected by traffic cam footage and aggregated right lane count values so:

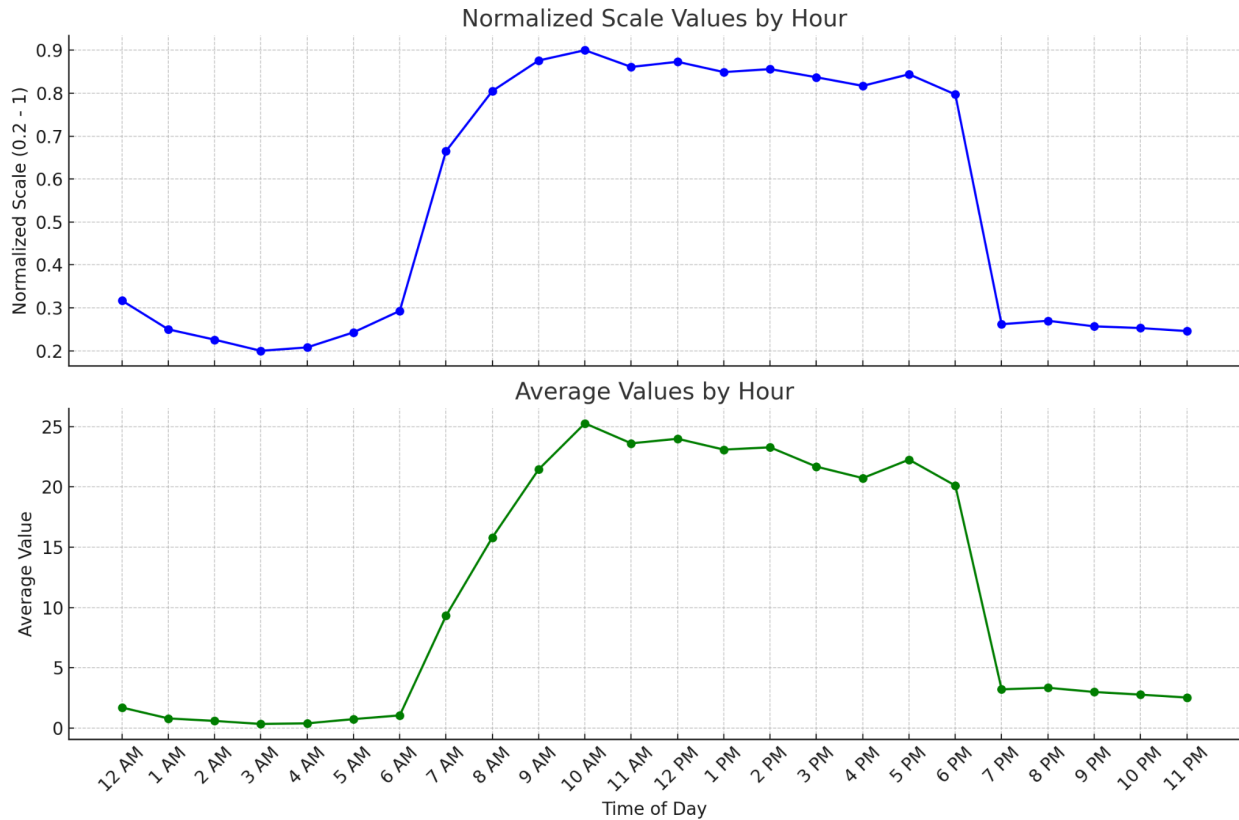
	A	B	C	D	E	F	G
1	Time	file_1	file_2	file_3	file_4	file_5	
2	00:00:00	2	3	1	1	1	
3	00:01:00	5	2	4	6	0	
4	00:02:00	0	1	1	6	3	
5	00:03:00	3	2	3	3	0	
6	00:04:00	1	0	3	2	2	
7	00:05:00	6	6	0	3	1	
8	00:06:00	0	0	3	1	1	
9	00:07:00	1	3	4	0	1	
10	00:08:00	5	2	6	9	1	
11	00:09:00	0	0	2	4	2	
12	00:10:00	3	4	2	0	3	
13	00:11:00	2	0	7	9	3	
14	00:12:00	3	4	0	2	0	
15	00:13:00	1	0	6	6	2	
16	00:14:00	3	5	1	2	0	
17	00:15:00	0	1	2	2	2	
18	00:16:00	2	1	3	1	0	
19	00:17:00	1	0	1	1	2	
20	00:18:00	0	4	2	2	0	
21	00:19:00	3	6	1	0	7	
22	00:20:00	0	4	3	6	0	
23	00:21:00	3	4	0	0	1	

Sampling to get avg per hour we can see the traffic demand curve

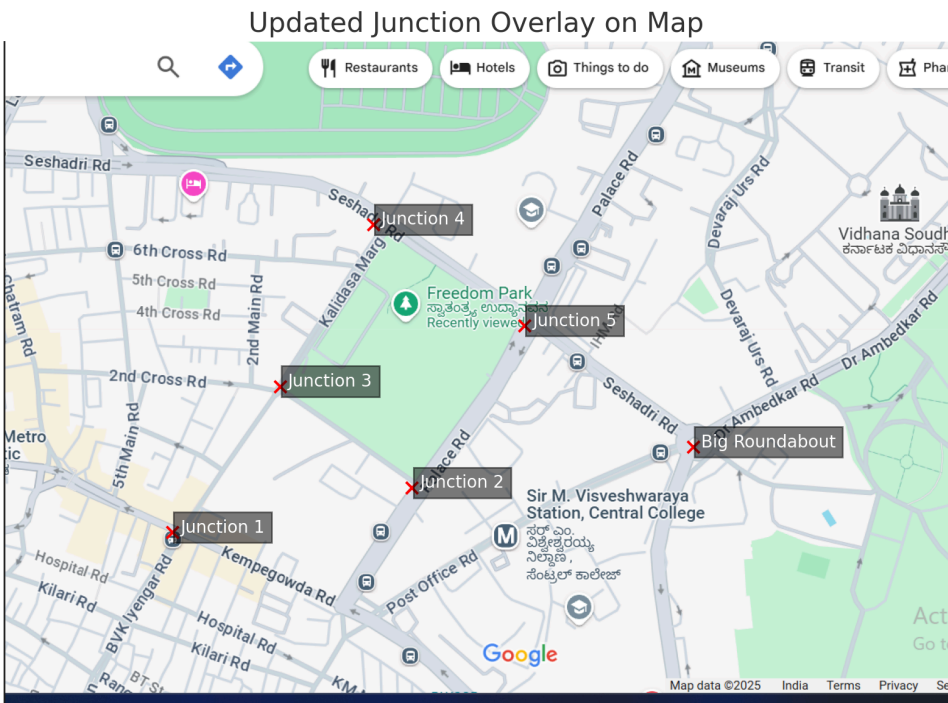


FREEDOM PARK 1.1 (Colab)

Normalized to get it between 0.2 and 1

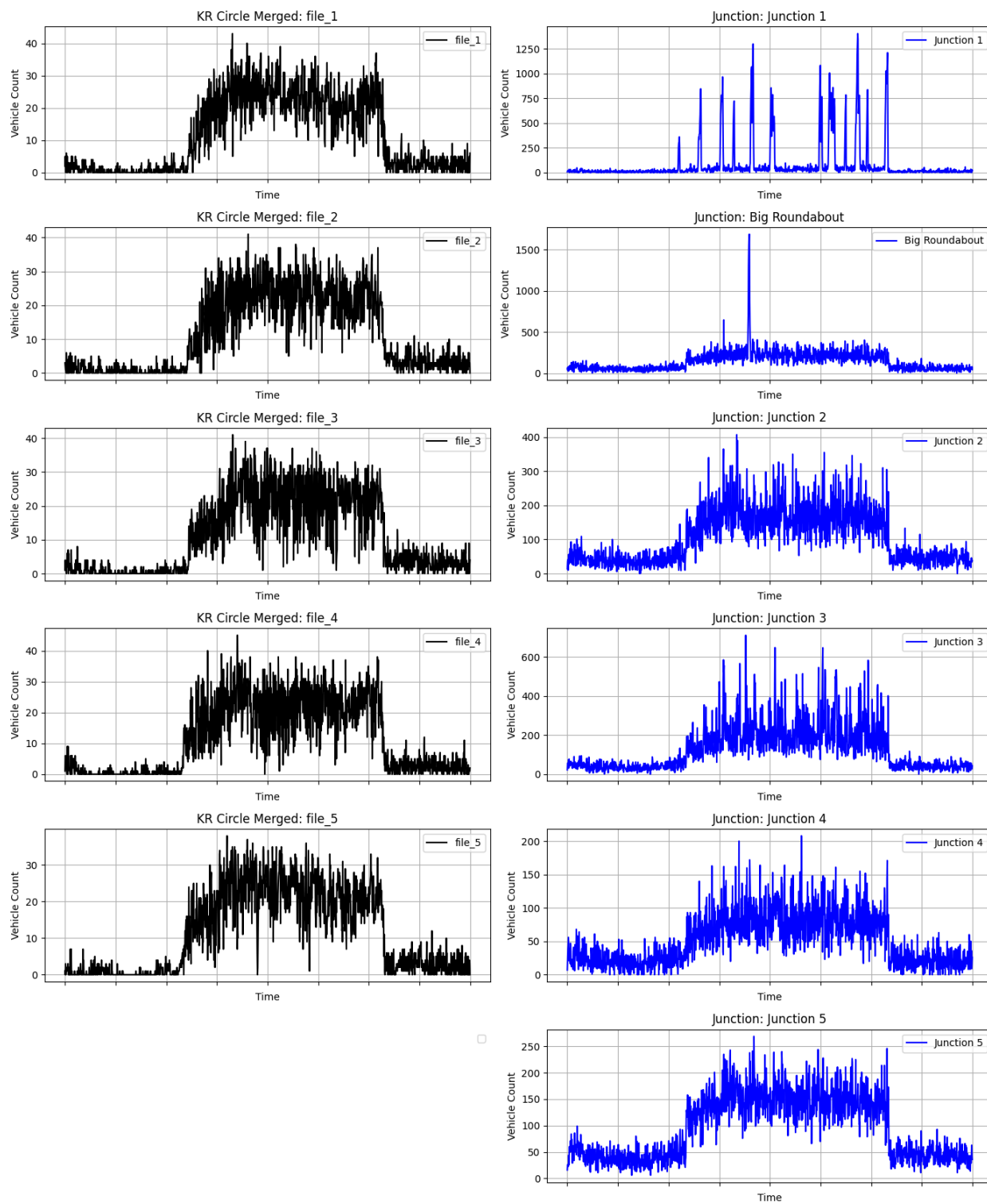


And used this to scale traffic in Traci



FREEDOM PARK 1.1 (Colab)

KR Circle Merged Data vs Junction Data (per Minute)



Left is KR Circle data

Right is Sumo Simulated data

FREEDOM PARK 1.1 (Colab)

Junction: Junction 1									
File	Cosine Similarity	Manhattan Distance	Euclidean Distance	Pearson Correlation	Spearman Correlation	RMSE	MAE	DTW Distance	
file_1	0.458605	103169	7886.352959	0.293789	0.604788	207.823648	71.645139	98652.0	
file_2	0.457261	103021	7885.971025	0.291398	0.607714	207.813583	71.542361	98211.0	
file_3	0.452234	103372	7898.063560	0.284078	0.609037	208.132250	71.786111	98968.0	
file_4	0.447102	103079	7890.908883	0.276727	0.615787	207.943707	71.582639	98714.0	
file_5	0.444219	103280	7898.497832	0.270901	0.598722	208.143694	71.722222	98685.0	
Junction: Big Roundabout									
File	Cosine Similarity	Manhattan Distance	Euclidean Distance	Pearson Correlation	Spearman Correlation	RMSE	MAE	DTW Distance	
file_1	0.832638	190824	6689.740354	0.641479	0.750979	176.290137	132.516667	189096.0	
file_2	0.830465	190692	6688.674308	0.635171	0.749765	176.262045	132.425000	189239.0	
file_3	0.820246	191361	6713.407406	0.614631	0.735558	176.913819	132.889583	189956.0	
file_4	0.822048	190706	6691.176877	0.620658	0.761465	176.327993	132.434722	188841.0	
file_5	0.840132	190939	6691.512011	0.654418	0.764623	176.336825	132.596528	189590.0	
Junction: Junction 2									
File	Cosine Similarity	Manhattan Distance	Euclidean Distance	Pearson Correlation	Spearman Correlation	RMSE	MAE	DTW Distance	
file_1	0.882233	134750	4368.838290	0.745152	0.753663	115.128998	93.576389	132108.0	
file_2	0.875338	134624	4370.919125	0.726373	0.744273	115.183833	93.488889	131824.0	
file_3	0.871258	135289	4392.804685	0.718605	0.751853	115.760568	93.950694	132679.0	
file_4	0.874980	134634	4367.965430	0.730234	0.762762	115.105996	93.495833	130893.0	
file_5	0.889446	134865	4370.969343	0.758494	0.762661	115.185156	93.656250	131934.0	
Junction: Junction 3									
File	Cosine Similarity	Manhattan Distance	Euclidean Distance	Pearson Correlation	Spearman Correlation	RMSE	MAE	DTW Distance	
file_1	0.852350	164511	5939.063478	0.691432	0.763683	156.508065	114.243750	159990.0	
file_2	0.849343	164377	5938.505283	0.683888	0.753074	156.493355	114.150694	159873.0	
file_3	0.838540	165054	5964.094567	0.662497	0.752971	157.167692	114.620833	162034.0	
file_4	0.841278	164389	5940.761652	0.669898	0.762016	156.552815	114.159028	158568.0	
file_5	0.850487	164626	5947.021776	0.684802	0.764883	156.717784	114.323611	160482.0	
Junction: Junction 4									
File	Cosine Similarity	Manhattan Distance	Euclidean Distance	Pearson Correlation	Spearman Correlation	RMSE	MAE	DTW Distance	
file_1	0.871812	57145	1878.829157	0.719792	0.716997	49.511496	39.684028	45685.0	
file_2	0.866266	56961	1880.696945	0.704391	0.708496	49.560716	39.556250	44875.0	
file_3	0.859245	57618	1903.593444	0.689617	0.716601	50.164092	40.012500	45388.0	
file_4	0.862731	56973	1880.506049	0.700651	0.725686	49.555686	39.564583	43000.0	
file_5	0.873925	57272	1883.654958	0.720528	0.730212	49.638667	39.772222	46380.0	
Junction: Junction 5									
File	Cosine Similarity	Manhattan Distance	Euclidean Distance	Pearson Correlation	Spearman Correlation	RMSE	MAE	DTW Distance	
file_1	0.904089	120614	3693.874389	0.813294	0.765573	97.342137	83.759722	118557.0	
file_2	0.896768	120482	3696.291926	0.790971	0.758728	97.405845	83.668056	118711.0	
file_3	0.889469	121149	3720.874225	0.773908	0.750749	98.053645	84.131250	119456.0	
file_4	0.893745	120496	3695.136804	0.788727	0.766788	97.375405	83.677778	118085.0	
file_5	0.906228	120727	3699.556595	0.813685	0.769041	97.491876	83.838194	119055.0	

Junction: Junction 1

File	Cosine Similarity	Manhattan Distance	Euclidean Distance	Pearson Correlation	Spearman Correlation	RMSE	MAE	DTW Distance
file_1	0.458605	103169	7886.352959	0.293789	0.604788	207.823648	71.645139	98652.0
file_2	0.457261	103021	7885.971025	0.291398	0.607714	207.813583	71.542361	98211.0
file_3	0.452234	103372	7898.063560	0.284078	0.609037	208.132250	71.786111	98968.0
file_4	0.447102	103079	7890.908883	0.276727	0.615787	207.943707	71.582639	98714.0

FREEDOM PARK 1.1 (Colab)

file_5	0.444219	103280	7898.497832	0.270901	0.598722	208.143694	71.722222	98685.0
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Junction: Big Roundabout

File	Cosine Similarity	Manhattan Distance	Euclidean Distance	Pearson Correlation	Spearman Correlation	RMSE	MAE	DTW Distance
file_1	0.832638	190824	6689.740354	0.641479	0.750979	176.290137	132.516667	189096.0
file_2	0.830465	190692	6688.674308	0.635171	0.749765	176.262045	132.425000	189239.0
file_3	0.820246	191361	6713.407406	0.614631	0.735558	176.913819	132.889583	189956.0
file_4	0.822048	190706	6691.176877	0.620658	0.761465	176.327993	132.434722	188841.0
file_5	0.840132	190939	6691.512011	0.654418	0.764623	176.336825	132.596528	189590.0

Junction: Junction 2

File	Cosine Similarity	Manhattan Distance	Euclidean Distance	Pearson Correlation	Spearman Correlation	RMSE	MAE	DTW Distance
file_1	0.882233	134750	4368.838290	0.745152	0.753663	115.128998	93.576389	132108.0
file_2	0.875338	134624	4370.919125	0.726373	0.744273	115.183833	93.488889	131824.0
file_3	0.871258	135289	4392.804685	0.718605	0.751853	115.760568	93.950694	132679.0
file_4	0.874980	134634	4367.965430	0.730234	0.762762	115.105996	93.495833	130893.0
file_5	0.889446	134865	4370.969343	0.758494	0.762661	115.185156	93.656250	131934.0

Junction: Junction 3

FREEDOM PARK 1.1 (Colab)

File	Cosine Similarity	Manhattan Distance	Euclidean Distance	Pearson Correlation	Spearman Correlation	RMSE	MAE	DTW Distance
file_1	0.852350	164511	5939.063478	0.691432	0.763683	156.508065	114.243750	159990.0
file_2	0.849343	164377	5938.505283	0.683888	0.753074	156.493355	114.150694	159873.0
file_3	0.838540	165054	5964.094567	0.662497	0.752971	157.167692	114.620833	162034.0
file_4	0.841278	164389	5940.761652	0.669898	0.762016	156.552815	114.159028	158568.0
file_5	0.850487	164626	5947.021776	0.684802	0.764883	156.717784	114.323611	160482.0

Junction: Junction 4

File	Cosine Similarity	Manhattan Distance	Euclidean Distance	Pearson Correlation	Spearman Correlation	RMSE	MAE	DTW Distance
file_1	0.871812	57145	1878.829157	0.719792	0.716997	49.511496	39.684028	45685.0
file_2	0.866266	56961	1880.696945	0.704391	0.708496	49.560716	39.556250	44875.0
file_3	0.859245	57618	1903.593444	0.689617	0.716601	50.164092	40.012500	45388.0
file_4	0.862731	56973	1880.506049	0.700651	0.725686	49.555686	39.564583	43000.0
file_5	0.873925	57272	1883.654958	0.720528	0.730212	49.638667	39.772222	46380.0

Junction: Junction 5

File	Cosine Similarity	Manhattan Distance	Euclidean Distance	Pearson Correlation	Spearman Correlation	RMSE	MAE	DTW Distance
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FREEDOM PARK 1.1 (Colab)

file_ 1	0.904089	120614	3693.8743 89	0.813294	0.765573	97.342 137	83.759 722	118557. 0
file_ 2	0.896768	120482	3696.2919 26	0.790971	0.758728	97.405 845	83.668 056	118711. 0
file_ 3	0.889469	121149	3720.8742 25	0.773908	0.750749	98.053 645	84.131 250	119456. 0
file_ 4	0.893745	120496	3695.1368 04	0.788727	0.766788	97.375 405	83.677 778	118085. 0
file_ 5	0.906228	120727	3699.5565 95	0.813685	0.769041	97.491 876	83.838 194	119055. 0